Project Roadmap

Team Name: SMU Peer Evaluation

Project Name: SMU Peer Evaluation System

Client: Singapore Management University (SMU)

Course: BIT 4454

Project Objective

Build a web-based peer evaluation system customized for SMU that allows faculty to manage evaluations and students to complete peer assessments, with analytics dashboards tied to SMU Graduate Learning Outcomes.

Measurable Objectives

Objective 1 (Sprint 1 – Student Peer Evaluation Submission)
Baseline: 0 peer evaluations submitted

Targets:
- At least 20 peer evaluations submitted per semester
- At least 80% of onboarded students submit a peer evaluation
- At least 90% of submitted peer evaluations include scores for at least 3 SMU Graduate Learning Outcomes

Objective 2 (Sprint 2 – Professor Scheduling)
Baseline: No scheduled evaluations

Targets:
- 100% of peer evaluations are associated with a scheduled evaluation record
- Each scheduled evaluation includes a course, due date, and professor

Objective 3 (Sprint 3 – Analytics & Reporting)
Baseline: No evaluation reports available

Targets:
- Administrators can view peer evaluation submission counts by course and date
- Administrators can view average peer evaluation scores by learning outcome

Product Backlog & Project Evaluation

Product Backlog (User Stories)

- As a Student, I need to submit a peer evaluation so that my contributions are assessed. (Sprint 1)
- As a Student, I need to select teammates to evaluate so that evaluations are correctly assigned. (Sprint 1)
- As a Professor, I need to schedule peer evaluations so that students know when to submit them. (Sprint 2)
- As a Professor, I need to import students and courses so that evaluations are associated with a class. (Sprint 3)

Team Project Evaluation Workbook

Inputs (Imports / User Actions):

- Peer Evaluation Submission *(Sprint 1)*
- Professor Schedule Peer Evaluation *(Sprint 2)*
- Student Import & Course Import *(Sprint 3)*

A	B	C	D
User Role	**User Story**	**Type**	**Sprint**
Student	Submit peer evaluations	Input	Sprint 1
Student	Select teammates to evaluate	Input	Sprint 1
Professor	Schedule peer evaluations	Input	Sprint 2
Administrator	View evaluation dashboards	Output	Sprint 3

Outputs (Reports / Analytics):

- **Peer Evaluation Submission Count (Bar Chart)**
  - X-axis: Course and submission date
  - Y-axis: Number of peer evaluations submitted
- **Average Peer Evaluation Score by Graduate Learning Outcome (Bar Chart)**
  - X-axis: Graduate Learning Outcome
  - Y-axis: Average score (1–4 scale)
- **Team Contribution Report (Table + Bar Chart)**
  - X-axis: Student
  - Y-axis: Average contribution score

Core Inputs

1. Peer Evaluation Submission (Sprint 1 - MVP)
    - Input type: Web form submission
    - Requirements:
        i. Student must select teammate to evaluate
        ii. Student enters rating
        iii. All fields must be completed
        iv. One submission per student per evaluation
        v. Submission is timestamped and recorded
2. Import Courses (Sprint 1 - MVP)
    - Input type: CSV file
    - Requirements:
        i. Must include course ID, course name, and term
3. Import Students (Sprint 1- MVP)
    - Input type: CSV file
    - Requirements:
        i. Student name, SMU email, and student ID
4. Team Assignment (Sprint 1- MVP)
    - Input type: Web form submission
    - Requirements:
        i. Students are assigned to teams

Core Outputs (BUSINESS INTELLIGENCE)

Course Peer Evaluation Scorecard

- For each course, there will be a card visual showing the average of contribution, communication, and collaboration.
- Management can flag courses with low averages. They can then review teamwork structure and the grading rubric.

Average Peer Evaluation Score per Professor

- There will be a bar chart of each professor displaying the average peer evaluation score.
- Management can identify professors who’s courses receive low scores and review expectation and guidance in courses taught by that professor. They can also implement practices of professors who’s courses receive high scores.

Course Peer Evaluation Distribution

- There will be a histogram showing the distribution of overall peer evaluation scores.
- Management can identify rating patterns and update peer evaluation guidance if scores are consistently clustered.

Peer Evaluation Completion

- There will be a table showing the amount of students enrolled and peer evaluations submitted per course.
- Management can identify with courses with low completion and take action, such as sending reminders, to improve student participation.

Visualization Tool: Tableau

Core Outputs (BUSINESS INTELLIGENCE)

1. Team Contribution Balance Dashboard *(Visual Report)*

- Identify teams with unbalanced participation levels across contribution, communication, and collaboration.
- Management action: Restructure teams or assign targeted teamwork coaching to improve balance.

2. Individual Performance Trend Report *(Visual Report)*

- Detect declining or improving peer-evaluation trends for individual students across evaluation cycles.
- Management action: Trigger academic advising, faculty check-ins, or recognize consistently high performers for leadership roles.

3. Rubric Dimension Effectiveness Table *(Tabular Report)*

- Identify rubric criteria that consistently score lower than others.
- Management action: Refine rubric definitions or adjust instructional emphasis on weak dimensions.

4. Course Comparison Performance Matrix *(Visual + Tabular Report)*

- Compare average peer-evaluation scores across courses and professors.
- Management action: Benchmark best-performing courses and apply high-impact instructional strategies to lower-performing courses.

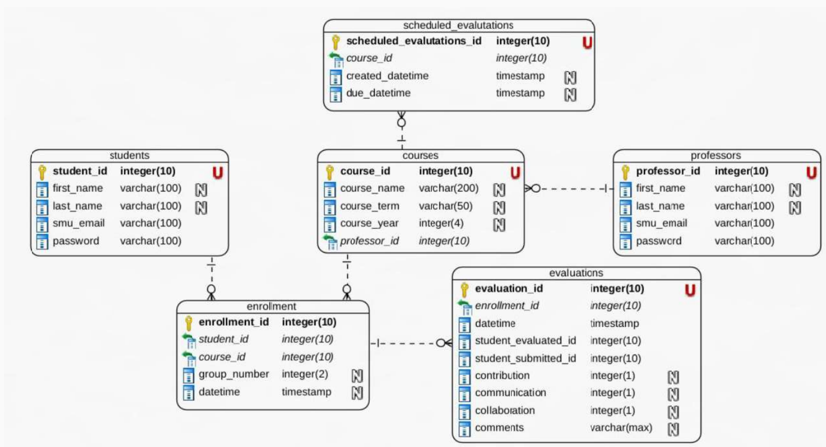
Visualization Tool: Power BI

Entity Relationship Diagram (DATA MANAGER)

Assumptions:

- All courses and professors also will have unique IDs. Professors will also have an SMU email.
- Professors can have multiples of the same course, but they must have separate IDs
- *evaluations* will store anonymous results, only storing the recipient of feedback.

Planned Database Platform: PostgreSQL



Sample Data (DATA MANAGER)

Initial Sample Data

professor_id	first_name	last_name	smu_email
2001	Robert	Anderson	robert.anderson@smu.edu.sg
2002	Jennifer	Lim	jennifer.lim@smu.edu.sg
2003	Michael	Tan	michael.tan@smu.edu.sg
2004	Susan	Chong	susan.chong@smu.edu.sg
2005	David	Kumar	david.kumar@smu.edu.sg
2006	Patricia	Wong	patricia.wong@smu.edu.sg
2007	James	Chen	james.chen@smu.edu.sg
2008	Linda	Ng	linda.ng@smu.edu.sg
2009	Christopher	Lee	christopher.lee@smu.edu.sg
2010	Michelle	Tan	michelle.tan@smu.edu.sg

student_id	first_name	last_name	smu_email
1001	Sarah	Chen	sarah.chen.2023@smu.edu.sg
1002	Marcus	Tan	marcus.tan.2023@smu.edu.sg
1003	Priya	Kumar	priya.kumar.2023@smu.edu.sg
1004	David	Lim	david.lim.2023@smu.edu.sg
1005	Amanda	Wong	amanda.wong.2023@smu.edu.sg
1006	Rajesh	Patel	rajesh.patel.2023@smu.edu.sg
1007	Emily	Ng	emily.ng.2023@smu.edu.sg
1008	Adrian	Koh	adrian.koh.2023@smu.edu.sg
1009	Melissa	Tan	melissa.tan.2023@smu.edu.sg
1010	Benjamin	Lee	benjamin.lee.2023@smu.edu.sg

evaluation_id	student_id	course_id	contribution	communication	collaboration	comments
4001	1001	3001	4	5	4	Excellent team player, always responsive
4002	1002	3001	5	4	5	Strong technical skills and helpful attitude
4003	1003	3001	3	4	3	Good contributions but could be more proactive
4004	1004	3001	5	5	5	Outstanding leader, facilitated great discussions
4005	1005	3001	4	3	4	Solid work, sometimes slow to respond
4006	1006	3002	5	5	4	Very organized and thorough in documentation
4007	1007	3002	4	4	5	Great collaborator, always willing to help
4008	1008	3002	3	3	3	Adequate performance, met basic requirements
4009	1009	3003	5	4	5	Innovative thinker, brought creative solutions
4010	1010	3003	4	5	4	Excellent communicator, kept team aligned

course_id	course_name	course_term	professor_id
3001	Database Management Systems	Spring2026	2001
3002	Software Engineering	Spring2026	2002
3003	Web Application Development	Spring2026	2003
3004	Data Analytics	Spring2026	2001
3005	Project Management	Spring2026	2004
3006	Database Management Systems	Spring2026	2001
3007	Mobile App Development	Spring2026	2005
3008	Business Intelligence	Spring2026	2002
3009	Cloud Computing	Spring2026	2006
3010	Cybersecurity Fundamentals	Spring2026	2007

Data Flow Diagram (PROCESS DESIGNER)

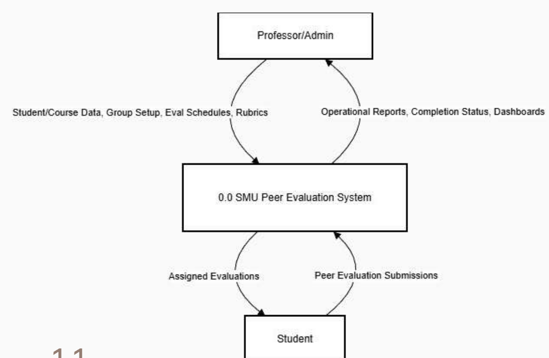
https://drive.google.com/file/d/13LXY02WrLFH1d6trAYbN8ta3thBARBoI/view?usp=sharing

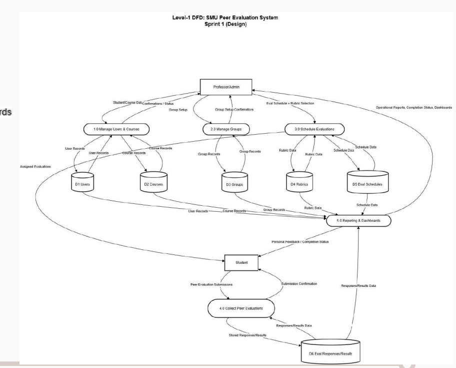
System Data Flow Diagram

- Level-1 DFD covering: imports → team creation → rubric setup → peer evaluation submission → reporting/analytics

Sprint Boundaries

- Sprint 1: manage users/courses + create groups + define rubrics (D1–D5)
- Sprint 2: peer evaluation submission + store responses (D6)
- Sprint 3: dashboards + reporting outputs





Crud Matrix (PROCESS DESIGNER)

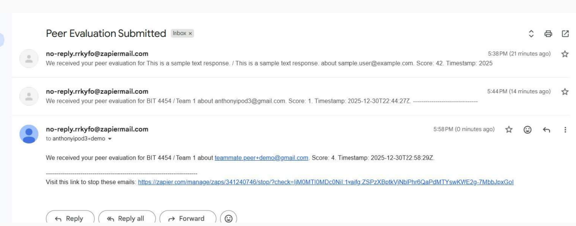
Process	Users	Courses	Groups/Teams	Rubrics	Eval Schedules	Peer Eval Submissions
1.0 Manage Users & Courses	C R U D	C R U D	R			
2.0 Manage Groups/Teams	R	R	C R U D			
3.0 Schedule Evaluations	R	R	R	C R U D	C R U D	
4.0 Collect Peer Evaluations	R	R	R	R	R	C R U D
5.0 Reporting & Dashboards	R	R	R	R	R	R

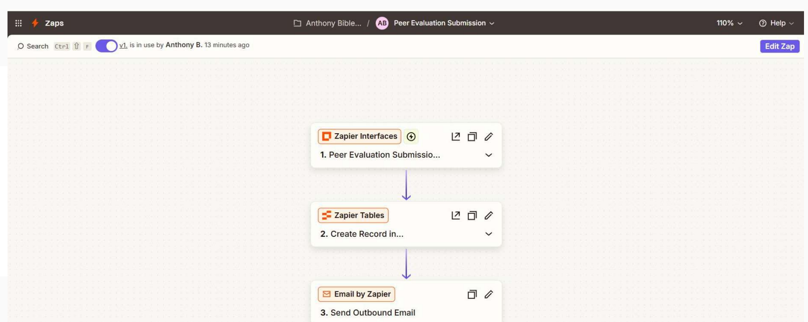
Process Integration Tool (PROCESS DESIGNER)

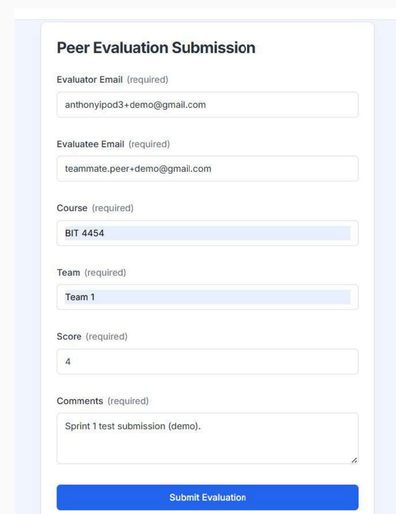
https://zapier.com/shared/2fcb99cb8c36445063eb85df1769b718f1853052

Process Integration Tool

- Tool Used: Zapier
  Usage: automated email reminders + deadline notifications + optional roster sync (CSV upload → database import trigger)
  Location: Cloud (Zapier platform)







UX Storyboards (UX LEAD)

User Experience Storyboards

Storyboards are illustrating the sequence of screens each stakeholder may encounter during their use of the core system tasks. The dashboards allows for a persistent and concise navigation to info screens including analytics and reports. These screens support the Professor & Admin input flows shown on the Data Flow Diagram and the project objectives.

System Used: JustInMind

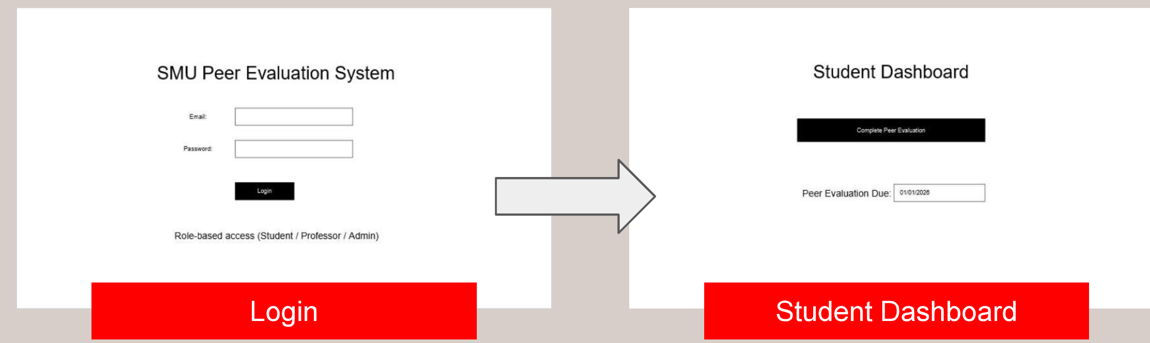
Location: Cloud (JustInMind Platform)

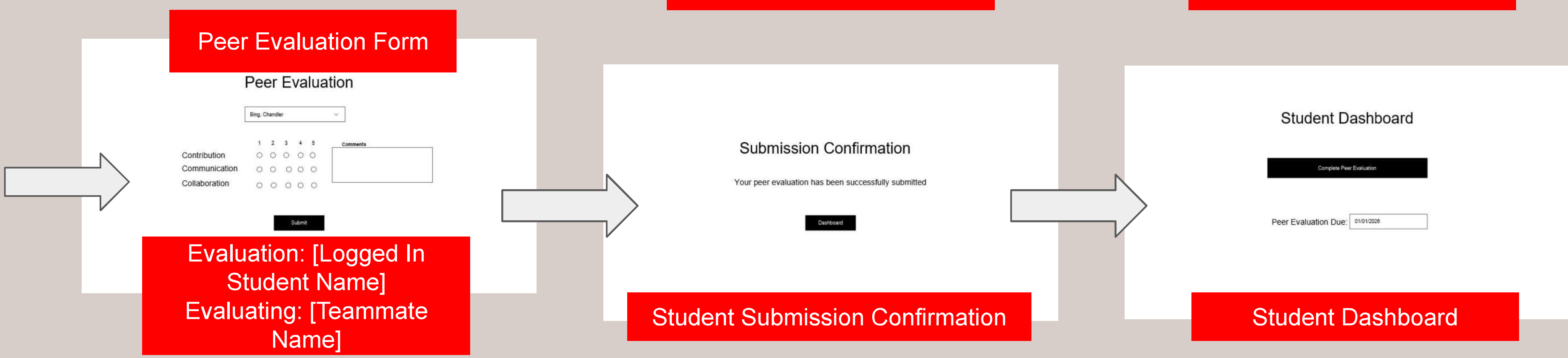
Student Storyboard - Sprint 1

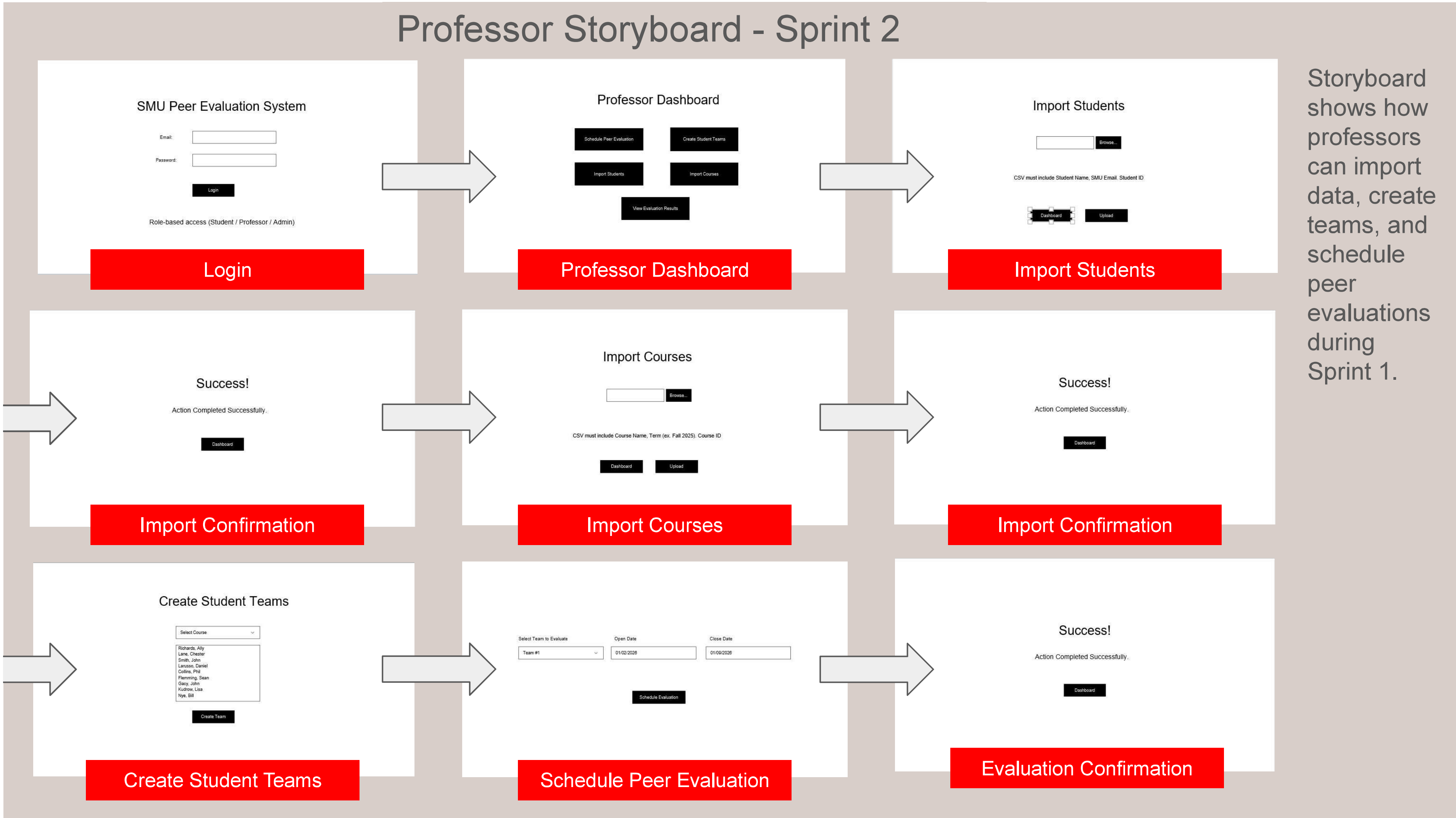
This storyboard demonstrates how students can access and submit assigned peer evaluations during Sprint 1. It delivers the core student peer evaluation submission workflow

Screens Implemented in Sprint 1:

- Student Dashboard
- Peer Evaluation
- Submission Confirmation



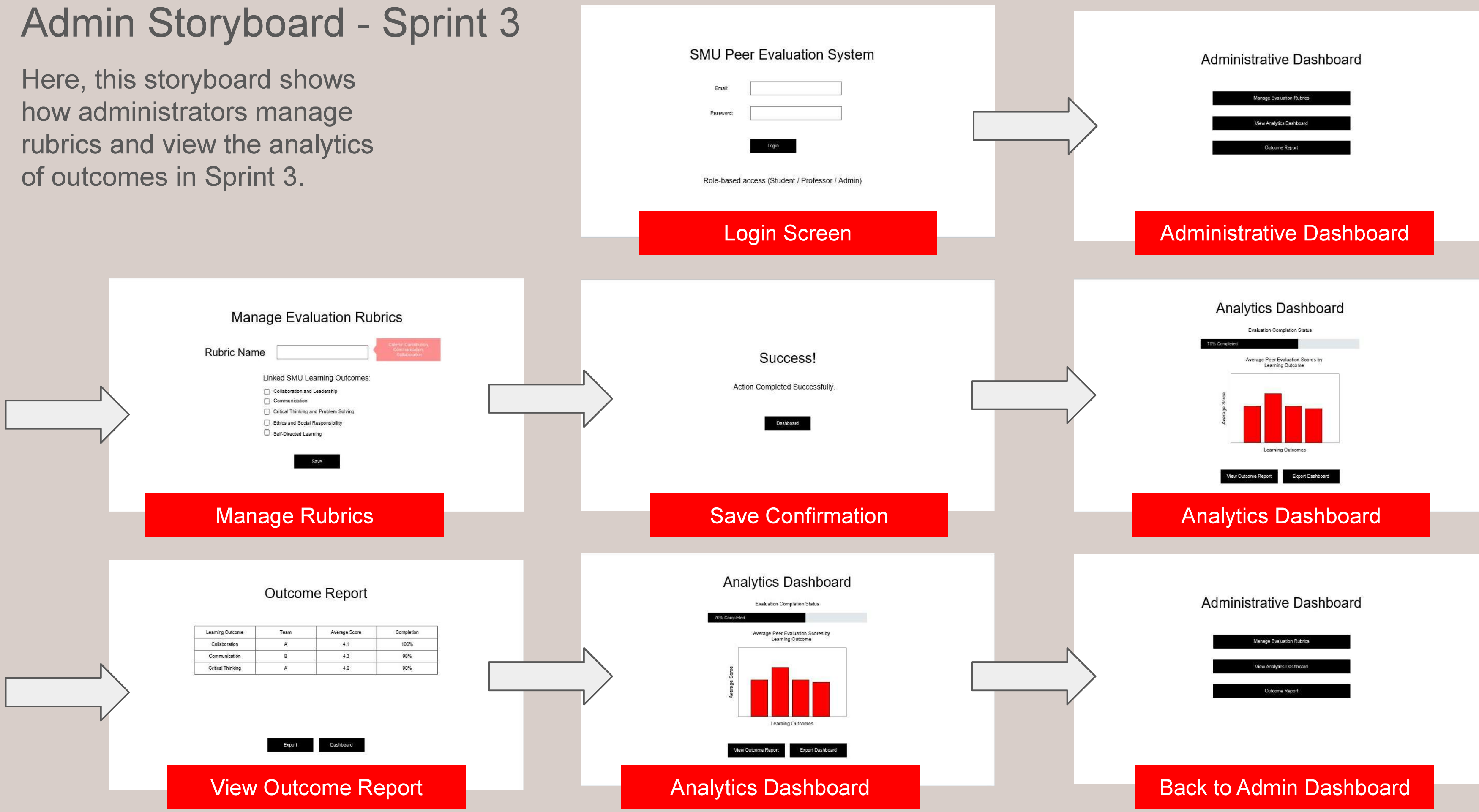




Professor Storyboard - Sprint 2
SMU Peer Evaluation System
Email:
Password:
Login
Role-based access (Student / Professor / Admin)
Login
Professor Dashboard
Schedule Peer Evaluation
Create Student Teams
Import Students
Import Courses
View Evaluation Results
Professor Dashboard
Import Students
Browse...
CSV must include Student Name, SMU Email, Student ID
Dashboard
Upload
Import Students
Success!
Action Completed Successfully.
Dashboard
Import Confirmation
Import Courses
Browse...
CSV must include Course Name, Term (ex. Fall 2025), Course ID
Dashboard
Upload
Import Courses
Success!
Action Completed Successfully.
Dashboard
Import Confirmation
Create Student Teams
Select Course
Richards, Ally
Lane, Chester
Smith, John
Larusso, Daniel
Collins, Phil
Flemming, Sean
Gacy, John
Kudrow, Lisa
Nye, Bill
Create Team
Create Student Teams
Select Team to Evaluate
Open Date
Close Date
Team #1
01/02/2026
01/09/2026
Schedule Evaluation
Schedule Peer Evaluation
Success!
Action Completed Successfully.
Dashboard
Evaluation Confirmation
Storyboard shows how professors can import data, create teams, and schedule peer evaluations during Sprint 1.

Admin Storyboard - Sprint 3

Here, this storyboard shows how administrators manage rubrics and view the analytics of outcomes in Sprint 3.



Technology Processes (Developer)

Professor/Admin (Student/Course Data, Group Setup, Eval Schedules, Rubrics):

- Frontend: Lovable, HTML, CSS
- Backend: Python, JavaScript
- Database: PostgreSQL

Student (Peer Evaluation Submissions):

- Input/Frontend: Lovable, HTML, CSS
- Backend: Python, JavaScript
- Data Storage: PostgreSQL

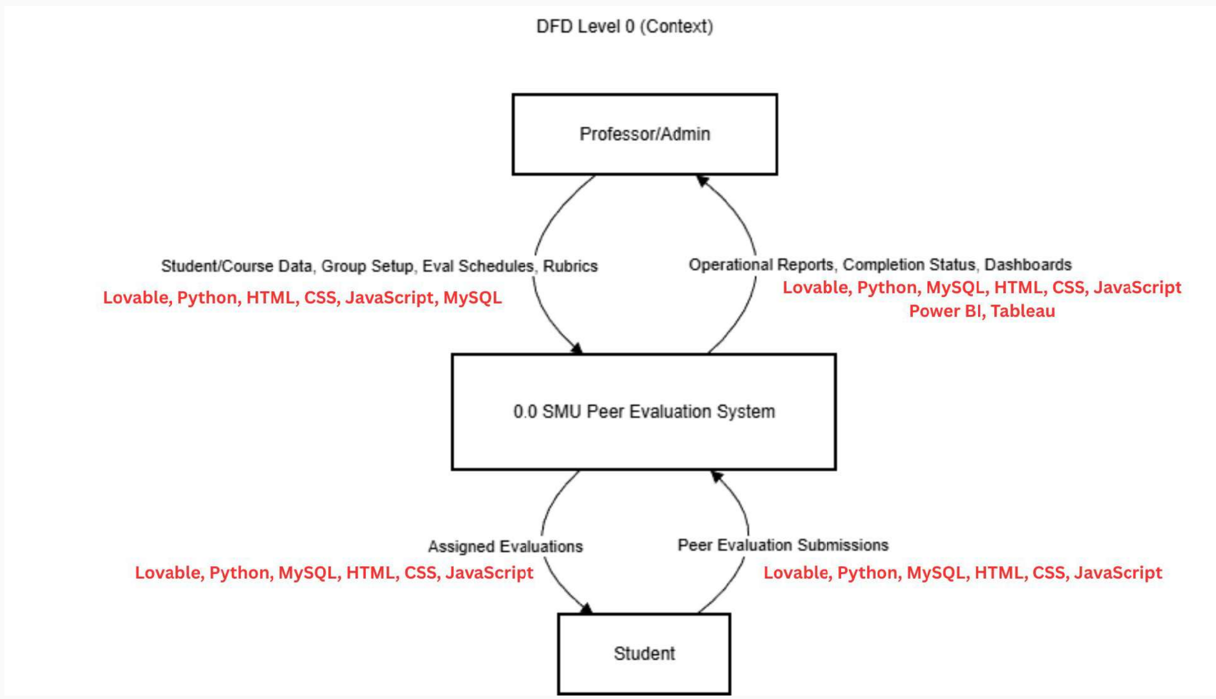
Student (Assigned Evaluations):

- Frontend: Lovable, HTML, CSS
- Backend: Python, JavaScript
- Database: PostgreSQL

SMU Peer Evaluation System (Operational Reports, Completion Status, Dashboards):

- Dashboard: JavaScript, Tableau, PowerBI
- Data Source: MySQL

Technology Processes (Developer)



Sprint 1 Minimum Viable Product

Sprint 1 will focus on:

- Student login
- View assigned peer evaluation
- Select teammate to evaluate
- Submit peer evaluation scores and comments
- Store peer evaluation submission with date/time

Out of Scope for Sprint 1

- Professor scheduling of evaluations
- Student and course imports
- Team creation and administration
- Analytics dashboards and reports